

$$77. \text{ Given exp.} = \frac{\frac{1}{3} \times 3 \times \frac{1}{3}}{\frac{1}{3} \div \frac{1}{9}} - \frac{1}{9} = \frac{\frac{1}{3}}{\frac{1}{3} \times 9} - \frac{1}{9} = \frac{1}{3} \times \frac{1}{3} - \frac{1}{9} = \frac{1}{9} - \frac{1}{9} = 0.$$

$$78. \text{ Given exp.} = \frac{\frac{1}{2} \div \frac{1}{4}}{\frac{1}{2} + \frac{1}{4}} = \frac{\frac{1}{2} \times 4}{\frac{2+1}{4}} = 2 \times \frac{4}{3} = \frac{8}{3} = 2\frac{2}{3}.$$

$$79. \text{ Given exp.} = \frac{\frac{13}{4} - \frac{4}{5} \text{ of } \frac{5}{6}}{\frac{13}{3} \div \frac{1}{5} - \left(\frac{3}{10} + \frac{106}{5}\right)} = \frac{\frac{13}{4} - \frac{2}{3}}{\frac{13}{3} \times 5 - \frac{215}{10}}$$

$$= \frac{\frac{31}{12} - \frac{43}{30}}{\frac{65}{3} - \frac{43}{2}} = \left(\frac{31}{12} \times 6\right) = \frac{31}{2} = 15\frac{1}{2}.$$

$$80. \text{ Let } \frac{\frac{15}{7} - \frac{23}{4}}{\frac{7}{2} + x} \div \frac{\frac{1}{6} + \frac{5}{7}}{\frac{5}{5} + \frac{2}{2}} = \frac{6}{10}.$$

$$\text{Then, } \left[\frac{7}{4} \times \frac{2}{(7+2x)}\right] \div \left[\frac{7}{4} \times \frac{10}{47}\right] = \frac{3}{5}$$

$$\Leftrightarrow \frac{7}{2(7+2x)} = \frac{3}{5} \times \frac{7}{4} \times \frac{10}{47} = \frac{21}{94}$$

$$\Leftrightarrow 7+2x = \left(\frac{7}{2} \times \frac{94}{21}\right) = \frac{47}{3}$$

$$\Leftrightarrow 2x = \frac{47}{3} - 7 = \frac{26}{3} \Leftrightarrow x = \left(\frac{26}{3} \times \frac{1}{2}\right) = \frac{13}{3} = 4\frac{1}{3}.$$

$$81. \frac{5}{3} + \frac{3}{4} = \frac{20+9}{12} = \frac{29}{12} = 2\frac{5}{12} < 5; \frac{7}{3} + \frac{11}{5}$$

$$= \frac{35+33}{15} = \frac{68}{15} = 4\frac{8}{15} < 5;$$

$$\frac{11}{4} + \frac{8}{3} = \frac{33+32}{12} = \frac{65}{12} = 5\frac{5}{12} > 5; \frac{13}{5} + \frac{11}{6}$$

$$= \frac{78+55}{30} = \frac{133}{30} = 4\frac{13}{30} < 5.$$

$$82. \text{ Given exp.} = 5 - \left[\frac{3}{4} + \left\{ \frac{5}{2} - \left(\frac{1}{2} + \frac{7-6}{42} \right) \right\} \right]$$

$$= 5 - \left[\frac{3}{4} + \left\{ \frac{5}{2} - \left(\frac{1}{2} + \frac{1}{42} \right) \right\} \right]$$

$$= 5 - \left[\frac{3}{4} + \left\{ \frac{5}{2} - \frac{22}{42} \right\} \right] = 5 - \left[\frac{3}{4} + \frac{83}{42} \right] = 5 - \frac{229}{84}$$

$$= \left(\frac{420-229}{84} \right) = \frac{191}{84} = 2\frac{23}{84}.$$

$$83. \text{ Given exp.} = -1 + 4 + 3 = 6.$$

$$84. x^3 - x^2 - x = (-3)^3 - (-3)^2 - (-3) = -27 - 9 + 3 = -33.$$

$$85. \text{ Putting } S = 8 \text{ and } R = 16, \text{ we have:}$$

$$R = gS - 4 \Leftrightarrow 16 = 8g - 4$$

$$\Leftrightarrow 8g = 20 \Leftrightarrow g = \frac{20}{8} = \frac{5}{2}.$$

$$\therefore \text{ When } S = 10, R = gS - 4$$

$$\Leftrightarrow R = \frac{5}{2} \times 10 - 4 = 25 - 4 = 21.$$

$$86. \frac{x(y-z)}{y(x+y+z)} = \frac{5(3-2)}{3(5+3+2)} = \frac{5}{3 \times 10} = \frac{1}{6}.$$

$$87. a^3 - b^3 + 3a^2b = 7^3 - 5^3 + 3 \times 7^2 \times 5$$

$$= 343 - 125 + 735 = 953.$$

$$88. \text{ Given exp.} = -7m - \{3n - \{8m - 4n + 10m\}\}$$

$$= -7m - \{3n - \{18m - 4n\}\}$$

$$= -7m - \{3n - 18m + 4n\} = -7m - \{7n - 18m\}$$

$$= -7m - 7n + 18m = 11m - 7n.$$

$$89. \frac{2}{a} + \frac{1}{b} = \frac{2b+a}{ab} = \frac{6}{4} = \frac{3}{2}.$$

$$90. \text{ Let } \frac{a}{3} = \frac{b}{4} = \frac{c}{7} = k. \text{ Then, } a = 3k, b = 4k, c = 7k.$$

$$\therefore \frac{(a+b+c)}{c} = \frac{3k+4k+7k}{7k} = \frac{14k}{7k} = 2.$$

$$91. \text{ Given exp.} = 2a^3 - \{3a^3 + 4b^3 - \{-5a^3\} + 5a^3 - 7b^3\}$$

$$= 2a^3 - \{3a^3 + 4b^3 + 5a^3 + 5a^3 - 7b^3\}$$

$$= 2a^3 - \{13a^3 - 3b^3\} = 2a^3 - 13a^3 + 3b^3$$

$$= -11a^3 + 3b^3.$$

$$92. \frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} = \frac{1}{1+\frac{x}{y+z}} + \frac{1}{1+\frac{y}{z+x}} + \frac{1}{1+\frac{z}{x+y}}$$

$$= \frac{y+z}{x+y+z} + \frac{z+x}{x+y+z} + \frac{x+y}{x+y+z}$$

$$= \frac{2(x+y+z)}{(x+y+z)} = 2.$$

$$93. (-2x-6)^2 = [-2(x+3)]^2 = 4(x+3)^2 = 4y.$$

$$94. (x+a)(x+b) = x^2 + ax + bx + ab = x^2 + (a+b)x + ab.$$

$$95. \left[1 + \frac{1}{x+1}\right] \left[1 + \frac{1}{x+2}\right] \left[1 + \frac{1}{x+3}\right] \left[1 + \frac{1}{x+4}\right]$$

$$= \left[\frac{(x+1)+1}{x+1}\right] \left[\frac{(x+2)+1}{x+2}\right] \left[\frac{(x+3)+1}{x+3}\right] \left[\frac{(x+4)+1}{x+4}\right]$$

$$= \left(\frac{x+2}{x+1}\right) \left(\frac{x+3}{x+2}\right) \left(\frac{x+4}{x+3}\right) \left(\frac{x+5}{x+4}\right) = \frac{x+5}{x+1}.$$

$$96. (a-b) - (c+d) = 6 \text{ and } (c-d) - (a+b) = 3$$

$$\Rightarrow (a-c) - (b+d) = 6 \text{ and } (c-a) - (b+d) = 3$$

$$\Rightarrow (b+d) = (a-c) - 6 \text{ and } (b+d) = (c-a) - 3$$

$$\Rightarrow (a-c) - 6 = (c-a) - 3 \Rightarrow 2(a-c) = 3$$

$$\Rightarrow (a-c) = \frac{3}{2} = 1.5.$$

$$97. \frac{3a+2b}{3a-2b} = \frac{3\frac{a}{b}+2}{3\frac{a}{b}-2} = \frac{3 \times \frac{4}{3} + 2}{3 \times \frac{4}{3} - 2} = \frac{6}{2} = 3.$$

$$98. \left(\frac{6}{7} - \frac{5x-y}{5x+y}\right) = \frac{6}{7} - \left(\frac{5\frac{x}{y}-1}{5\frac{x}{y}+1}\right) = \frac{6}{7} - \left(\frac{5 \times \frac{6}{5} - 1}{5 \times \frac{6}{5} + 1}\right)$$

$$= \frac{6}{7} - \left(\frac{6-1}{6+1}\right) = \frac{6}{7} - \frac{5}{7} = \frac{1}{7}.$$